When constant g is not enough

antocciphysics | TEACHER KEY | CPA

Learning target: Recognize the limits of the near-surface mgh model.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 5; 3 course days later.*
22 J, using $K_f=K_i+W_{net}$.

02. Explain a worked example (worked)
Field changes with distance; surface-g approximation is poor over planet-scale heights.

03. Look and explain (model)
Book lift; 1 m is tiny relative to radius. Satellite field changes strongly.

04. Build the solution (practice)
Overestimate; actual gravitational pull decreases along path.

05. Use the idea (reason)
r=R+h; using h ignores radius and gives wrong field/energy.

06. Choose a method (mixed) *Review: Unit 3, Day 10; 12 course days later.*
6 m/s west and 8 m/s south.

07. Exit check (exit)
Constant g; height change tiny relative to Earth radius.

08. STRETCH: Apply it another way (stretch)
Small altitude is closer to constant 12; at h=R, field 3 N/kg.

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Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 5; 3 course days later.*
Net work 37.5 J; v_f=5 m/s from ½mv²=W_{net}.

02. Explain a worked example (worked)
Field changes with distance; surface-g approximation is poor over planet-scale heights.

03. Represent and interpret (model)
Book lift; 1 m is tiny relative to radius. Satellite field changes strongly.

04. Solve and justify (practice)
Exact 10 J; constant g estimate 20 J; overestimates.

05. Analyze the reasoning (reason)
r=R+h; using h ignores radius and gives wrong field/energy.

06. Choose a method (mixed) *Review: Unit 3, Day 10; 12 course days later.*
Other p=(20, -6) kg·m/s; magnitude≈20.88 kg·m/s.

07. Exit check (exit)
1 and .25 N/kg; strong variation invalidates constant g over this interval.

08. STRETCH: Apply it another way (stretch)
Exact 200(1/5-1/10)=20 J. Surface g=4; mgh=2×4×5=40 J; approximation overestimates.